



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Name : Vaibhav Sinha

Registration Number : 23BDS1026

Slot : L23+L24

Course : Exploratory Data Analysis Lab

Experiment 5

Code

```
house <- read.csv("train.csv", stringsAsFactors = FALSE)
```

```
str(house)
```

```
summary(house)
```

```
missing_count <- colSums(is.na(house))
```

```
missing_attributes <- missing_count[missing_count > 0]
```

```
cat("ATTRIBUTES WITH MISSING VALUES\n")
```

```
print(missing_attributes)
```

```
missing_percent <- (missing_count / nrow(house)) * 100
```

```
missing_percent_df <- data.frame(
```

```
  Attribute = names(missing_percent),
```

```
Missing_Percentage = round(missing_percent, 2)
)
```

```
cat("PERCENTAGE OF MISSING VALUES\n")
print(missing_percent_df[missing_percent_df$Missing_Percentage > 0, ])
```

```
numeric_cols <- sapply(house, is.numeric)
```

```
numeric_missing <- names(house)[numeric_cols &
                                colSums(is.na(house)) > 0]
```

```
selected_cols <- head(numeric_missing, 5)
```

```
cat("SELECTED NUMERICAL ATTRIBUTES\n")
print(selected_cols)
```

```
before_data <- house[selected_cols]
```

```
house_imputed <- house
```

```
for(col in selected_cols) {
  mean_val <- mean(house_imputed[[col]], na.rm = TRUE)
  house_imputed[[col]][is.na(house_imputed[[col]])] <- mean_val
}
```

```
comparison <- data.frame(
```

```

Attribute = character(),
Mean_Before = numeric(),
SD_Before = numeric(),
Mean_After = numeric(),
SD_After = numeric(),
stringsAsFactors = FALSE
)

for(col in selected_cols) {

  mean_before <- mean(house[[col]], na.rm = TRUE)
  sd_before <- sd(house[[col]], na.rm = TRUE)

  mean_after <- mean(house_imputed[[col]])
  sd_after <- sd(house_imputed[[col]])

  comparison <- rbind(
    comparison,
    data.frame(
      Attribute = col,
      Mean_Before = round(mean_before, 2),
      SD_Before = round(sd_before, 2),
      Mean_After = round(mean_after, 2),
      SD_After = round(sd_after, 2)
    )
  )
}

```

```
}
```

```
cat("MEAN & SD COMPARISON\n")
```

```
print(comparison)
```

```
cat("FIRST 10 RECORDS BEFORE IMPUTATION\n")
```

```
print(before_data[1:10, ])
```

```
cat("FIRST 10 RECORDS AFTER IMPUTATION\n")
```

```
print(house_imputed[selected_cols][1:10, ])
```

Output

```
> source("../active-rstudio-document")
"data.frame": 1460 obs. of 81 variables:
 $ id          : int  1 2 3 4 5 6 7 8 9 10 ...
 $ MSSubclass  : int  60 20 60 70 60 50 20 60 50 190 ...
 $ MSZoning    : chr   "RL" "RL" "RL" ...
 $ LotFrontage : int  65 80 68 60 84 85 75 NA 51 50 ...
 $ LotArea     : int  8450 9600 11250 9550 14260 14115 10084 10382 6120 7420 ...
 $ Street      : chr   "Pave" "Pave" "Pave" "Pave" ...
 $ Alley       : chr   NA NA NA NA ...
 $ LotShape    : chr   "Reg" "Reg" "IR1" "IR1" ...
 $ LandContour : chr   "Lvl" "Lvl" "Lvl" "Lvl" ...
 $ Utilities   : chr   "Allpub" "Allpub" "Allpub" "Allpub" ...
 $ LotConfig   : chr   "Inside" "IR2" "Inside" "Corner" ...
 $ LandSlope   : chr   "Gtl" "Gtl" "Gtl" "Gtl" ...
 $ Neighborhood : chr   "Collgr" "veenker" "collgr" "Crawfor" ...
 $ Condition1  : chr   "norm" "Feedr" "norm" "norm" ...
 $ Condition2  : chr   "norm" "norm" "norm" "norm" ...
 $ BldgType    : chr   "1fam" "1fam" "1fam" "1fam" ...
 $ HouseStyle  : chr   "2story" "1story" "2story" "2story" ...
 $ OverallQual : int  7 6 7 7 8 5 8 7 5 ...
 $ OverallCond : int  5 8 5 5 5 5 5 6 5 6 ...
 $ YearBuilt   : int  2003 1976 2001 1915 2000 1993 2004 1973 1931 1939 ...
 $ YearRemoddd : int  2003 1976 2002 1970 2000 1993 2005 1973 1950 1900 ...
 $ RoofStyle   : chr   "Gable" "Gable" "Gable" "Gable" ...
 $ RoofMatl    : chr   "Compshg" "Compshg" "Compshg" "Compshg" ...
 $ Exterior1st : chr   "VinylSd" "MetalSd" "VinylSd" "wd shng" ...
 $ Exterior2nd : chr   "VinylSd" "MetalSd" "VinylSd" "wd shng" ...
 $ MasVnrType  : chr   "BrkFace" "None" "BrkFace" "None" ...
 $ MasVnrArea  : int  196 0 162 0 310 0 166 240 0 0 ...
 $ ExterQual   : chr   "Gd" "TA" "Gd" "TA" ...
 $ ExterCond   : chr   "TA" "TA" "TA" "TA" ...
 $ Foundation  : chr   "PconC" "Cblock" "pconC" "BrkTill" ...
 $ BsmtQual    : chr   "Gd" "Gd" "Gd" "TA" ...
 $ BsmtCond    : chr   "TA" "TA" "TA" "Gd" ...
 $ BsmtExposure : chr   "No" "Gd" "No" "No" ...
 $ BsmtFinType1 : chr   "GLQ" "ALQ" "GLQ" "ALQ" ...
 $ BsmtFinS1   : int  706 978 486 216 655 732 1369 859 0 851 ...
 $ BsmtFinType2 : chr   "Unf" "Unf" "Unf" "Unf" ...
 $ BsmtFinS2   : int  0 0 0 0 0 0 32 0 0 ...
 $ BsmtFinSF1  : int  150 284 438 540 490 64 317 216 952 340 ...
 $ TotalBsmtSF : int  856 1260 920 756 1145 796 1686 1107 952 991 ...
 $ Heating     : chr   "GasA" "GasA" "GasA" "GasA" ...
 $ HeatingCov : chr   "Ex" "Ex" "Ex" "Gd" ...
 $ CentralAir  : chr   "Y" "Y" "Y" "Y" ...
 $ Electrical  : chr   "Sbrkr" "Sbrkr" "Sbrkr" "Sbrkr" ...
 $ Misc1stFlr : int  856 1262 920 963 1145 796 1694 1107 1022 1077 ...
 $ Misc2ndFlr : int  854 0 846 756 1013 566 0 983 757 0 ...
 $ LowQualFinS : int  0 0 0 0 0 0 0 0 0 ...
 $ GrLivArea   : int  1710 1282 1786 1717 2198 1362 1694 2090 1774 1077 ...
 $ BsmtHalfBth : int  1 0 1 1 1 1 1 0 1 ...
 $ BsmtHalfBth : int  0 1 0 0 0 0 0 0 0 ...
 $ FullBath    : int  2 2 2 1 2 1 2 2 2 ...
 $ HalfBath    : int  1 0 1 0 1 1 0 1 0 ...
 $ BedroomAbvGr : int  3 3 3 4 1 3 3 2 2 ...
 $ KitchenAbvGr : int  1 1 1 1 1 1 1 2 2 ...
 $ KitchenQual : chr   "Gd" "TA" "Gd" "Gd" ...
```

```

      81      81      81      81      1453
MiscFeature
1406
PERCENTAGE OF MISSING VALUES
      Attribute Missing_Percentage
LotFrontage LotFrontage 17.74
Alley Alley 93.77
MasVnrType MasVnrType 0.55
MasVnrArea MasVnrArea 0.55
BsmtQual BsmtQual 2.53
BsmtCond BsmtCond 2.53
BsmtExposure BsmtExposure 2.60
BsmtFinType1 BsmtFinType1 2.53
BsmtFinType2 BsmtFinType2 2.60
Electrical Electrical 0.07
FireplaceQu FireplaceQu 47.26
GarageType GarageType 5.55
GarageYrBlt GarageYrBlt 5.55
GarageFinish GarageFinish 5.55
GarageQual GarageQual 5.55
GarageCond GarageCond 5.55
PoolQC PoolQC 99.52
Fence Fence 80.75
MiscFeature MiscFeature 96.30
SELECTED NUMERICAL ATTRIBUTES
[1] "LotFrontage" "MasVnrArea" "GarageYrBlt"
MEAN & SD COMPARISON
      Attribute Mean_Before SD_Before Mean_After SD_After
1 LotFrontage 70.05 24.28 70.05 22.02
2 MasVnrArea 103.69 181.07 103.69 180.57
3 GarageYrBlt 1978.51 24.69 1978.51 23.99
FIRST 10 RECORDS BEFORE IMPUTATION
      LotFrontage MasVnrArea GarageYrBlt
1 65 196 2003
2 80 0 1976
3 68 162 2001
4 60 0 1998
5 84 350 2000
6 85 0 1993
7 75 186 2004
8 NA 240 1973
9 51 0 1931
10 50 0 1939
FIRST 10 RECORDS AFTER IMPUTATION
      LotFrontage MasVnrArea GarageYrBlt
1 65.00000 196 2003
2 80.00000 0 1976
3 68.00000 162 2001
4 60.00000 0 1998
5 84.00000 350 2000
6 85.00000 0 1993
7 75.00000 186 2004
8 70.04996 240 1973
9 51.00000 0 1931
10 50.00000 0 1939
> |

```